



"Your Partner in Innovative Refrigeration and Advanced Testing Solutions."

BATTERY TEST CHAMBERS



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BATTERY TEST CHAMBERS

ENSURING THE RELIABILITY AND SAFETY OF YOUR ENERGY STORAGE SYSTEMS

At **Sri Easwari Scientific Solution Pvt Ltd (SESS)**, we are committed to providing advanced testing solutions for the rapidly evolving energy storage industry. Our Battery Test Chambers are designed to meet the stringent requirements for testing lithium-ion and other energy storage systems, ensuring safety, reliability and performance.



Usage Areas and Applications:

- **Electric Vehicles (EVs)**

Test battery packs and modules for durability, safety, and performance.

- **Renewable Energy Storage**

Validate the reliability and efficiency of batteries used in solar and wind energy systems.

- **Consumer Electronics**

Ensure the safety and longevity of batteries in smartphones, laptops, and other electronic devices.

- **Industrial Applications**

Test batteries used in industrial machinery and equipment for consistent performance under various conditions.

- **International Standards Covered**

IEC 62660 , UL 1642, UN 38.3 , SAE J2464 , ISO 12405

Advantages of Testing with ESS Battery Test Chambers:

● Safety Assurance

Simulate extreme conditions to ensure battery safety, including thermal runaway and fire propagation tests.

● Performance Validation

Test battery capacity, cycle life, and efficiency under different environmental conditions.

● Precise Control

Advanced PLC control systems with HMI provide accurate and repeatable test results.

● Comprehensive Data Logging

Extensive data logging capabilities for detailed analysis and reporting.

● Customization

Chamber can be tailored to meet specific testing requirements and protocols.

● Enhanced Reliability

Identify potential issues early in the development process to improve battery design and performance.

Technical Specification	
SERIES	: ETC – Climatic Test Chamber
PERFORMANCE	: Battery test chamber designed for steady-state and cyclic tests.
WORK SPACE	: Capacity : 180, 250, 340, 380, 500, 600... 1000 liters (customized sizes available upon request).
TEMPERATURE RANGE	: -20°C to 120°C / -40°C to 120°C / -70°C to 120°C. Control Accuracy : ±1°C. Temperature Uniformity/Gradient : ±2°C.
HUMIDITY	: RH : 10% to 95%. Control Accuracy: ±3%.
CONDENSER TYPE	: A / W: "A" for Air-cooled, "W" for Water-cooled. Low-noise refrigeration system. Up to -40°C single-stage and -70°C double-stage Bitzer/Dorin compressor. CFC-free refrigerant (R404A).
HEATING RATE OF CHANGE OF TEMPERATURE.	: H 3 : 3 °C /min (Empty chamber), Average Transition Rate from Ambient (°C/minute) : As per IEC 60068-3-5, Temp from -40 °C to 120 °C with sensor in discharge of air of blower.
COOLING RATE OF CHANGE OF TEMPERATURE.	: C 3 : 3 °C /min (with 1 ton mass) Average Transition Rate from Ambient (°C/minute) : As per IEC 60068-3-5, Temp from 120 °C to -40°C with sensor in discharge of air of blower.
HAZARD LEVEL GAS FORMATION	: Hazard Level : Up to 7 (based on customer request). Gas Formation : CO, O2, CO2, H2
CONTROLLER FEATURES	: PLC-based control system. PLC: Siemens S7-1200. HMI: 7-inch Weintek display. Optional PC software: NI-based PC software. SESS Climatic Viewer, developed by SESS. Internal Memory: 4 GB. Data Logging: Up to 90 days (based on sampling duration, minimum 10 seconds). RAM Memory: 1 GB. USB Port: 1. Ethernet Ports: 2. RS 485 Port: 1. Optional PC software: SESS Climatic Test Viewer (LabVIEW-based software).

Battery Test Chambers: Key Safety Features

Hazard Level: No Risk of Explosive Atmosphere in the Chamber

Our Battery Test Chambers are engineered to ensure the highest level of safety during testing. One of the critical features is the hazard level, designed specifically to mitigate any risk of an explosive atmosphere within the chamber. This ensures a safe testing environment, even under extreme conditions.

Key safety components include

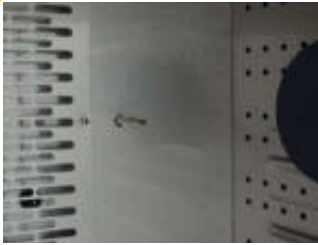
Optical/Acoustic Alarm

This system provides immediate hazard notifications through both visual and auditory alerts. The optical alarm uses a high-intensity LED beacon or strobe light to ensure visibility, while the acoustic alarm employs a loud buzzer or siren to grab attention. These alarms are activated upon detection of hazardous conditions such as gas leaks, overpressure, or extreme temperatures, ensuring prompt response and enhanced safety.



Supplementary PT100 Sensor

The PT100 sensor is a precision temperature sensor that provides additional temperature monitoring within the test chamber. It offers accurate readings and quick response times, ensuring that temperature variations are detected and managed effectively, thus maintaining the integrity of the testing environment and preventing overheating or other temperature-related issues.



Reversible Overpressure Valve

This valve is designed to manage pressure changes safely within the chamber. It automatically releases excess pressure, preventing potential damage to the chamber or hazardous situations. The valve can reset itself after activation, ensuring continuous protection without the need for manual intervention.



Additional Door Locks

Enhanced security is provided through additional door locks, which ensure that the chamber remains sealed during operation. These locks prevent unauthorized access and maintain a controlled environment, which is crucial for the accuracy and safety of the tests being conducted.

Battery Gas Monitoring

Continuous monitoring of hydrogen (H₂) and carbon monoxide (CO) levels is critical in battery testing environments. This system uses advanced sensors to detect the presence of these gases, which can be indicative of potentially dangerous conditions. Immediate alerts are provided if gas levels exceed safe thresholds, allowing for prompt corrective action.



Compressed Air or GN2 Washing

This feature maintains a safe atmosphere within the chamber by using compressed air or gaseous nitrogen (GN₂) to purge and clean the internal environment. This process helps to remove any contaminants or hazardous gases, ensuring a controlled and safe testing space.

Porthole with Retaining Clamps

Secure access points are provided through portholes with retaining clamps. These portholes allow for the introduction or removal of test samples without compromising the chamber's environment.



Door Locking System

The door locking system prevents unauthorized access during testing. It ensures that the chamber remains sealed, protecting both the test samples and personnel. This system is critical for maintaining the integrity of the tests and preventing accidental exposure to hazardous conditions.



CO2 Protection System/Water Mist Predisposition



This system ensures a rapid response in case of fire. The CO2 protection system can quickly flood the chamber with carbon dioxide to extinguish flames, while the water mist predisposition allows for the integration of a water mist system. Both methods provide effective fire suppression, protecting both the equipment and personnel.

These features collectively ensure that our Battery Test Chambers provide a secure and controlled environment for all your battery testing needs, adhering to the highest safety standards.

Refrigerant

Type: Single-stage / Double-stage compressor, **Refrigerant:** R404A / R23

Control System

PLC: Siemens S7-1200 , **HMI:** 7-inch Weintek, **Sensor:** Temp RTD Pt100 Class A,

Display Resolution: 0.01°C, **Communication:** LAN 1 & 2 on HMI, USB port for data download.

Construction Features:

Interior: Mirror-finish stainless steel sheets (SS 304), thickness 1.5/1.2 mm, TIG-welded

Exterior: CRCA sheet thickness 1.5 mm, Powder coated

Insulation Thickness: 125 mm glass wool

Electrical Panel

Design: Accommodates all necessary switchgear items

Switchgear Items: Separate MCBs, contactors, OLR/MPCBs for fan motors and refrigeration compressors

Additional Features: MCBs, contactors, separate SSRs for each phase for air heaters

Control Panel Box: Proper ferrule numbering and wiring

Port Hole

Size: 75 mm

Tray:

Type: 4 mm rod mesh

Chamber Inner Viewing Lamp:

Power: 20 watts

Location: Viewing window glass area

Control: On/Off via HMI

Door

Design: Top opening, double-walled insulated

Materials: Outer CRCA sheet, interior stainless steel construction

Features: Equipped with a limit switch, positioned on the right side of the door

Features

Illumination: White LED illumination at suitable locations ,

Heater: ensure clear viewing even at temperatures 10°C lower than test conditions,

Defrost heater: Provision available

Door Gasket

Design: Double-layer silicon rubber, **Placement:** One gasket on the chamber, one on the door

Hinges & Latches

Hinges: Heavy duty, **Latches:** Toggle type locking arrangement

Mounting

Type: Vertical floor-mount

Caster Wheels: With isolation jack option

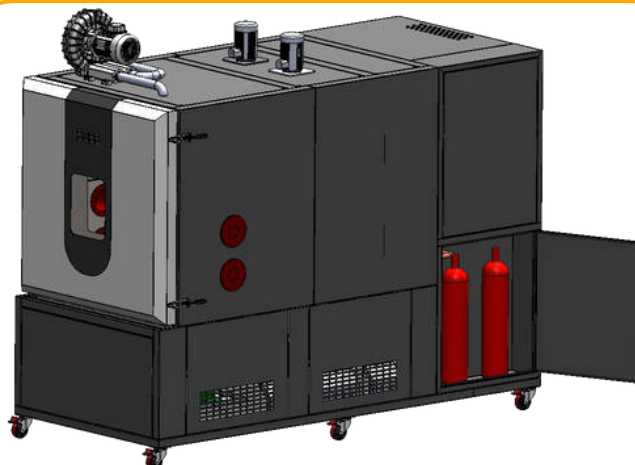
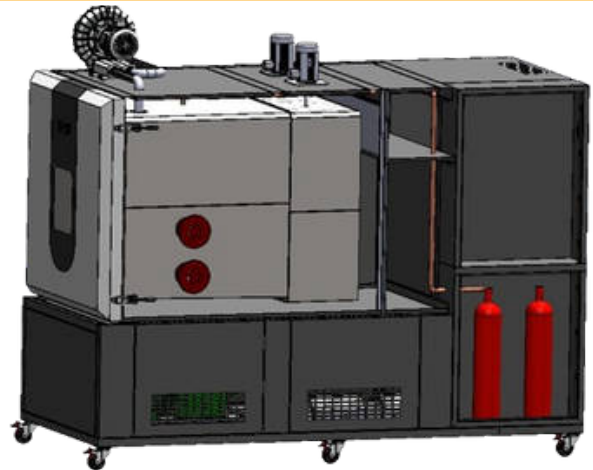
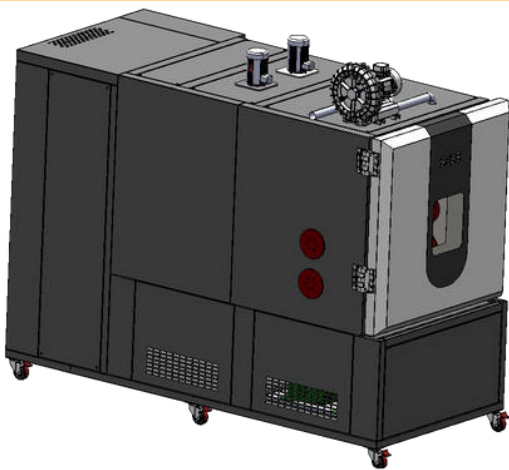
Quantity: Four heavy-duty, lockable, non-metallic, **Capacity:** 600 kg to 800 kg per wheel

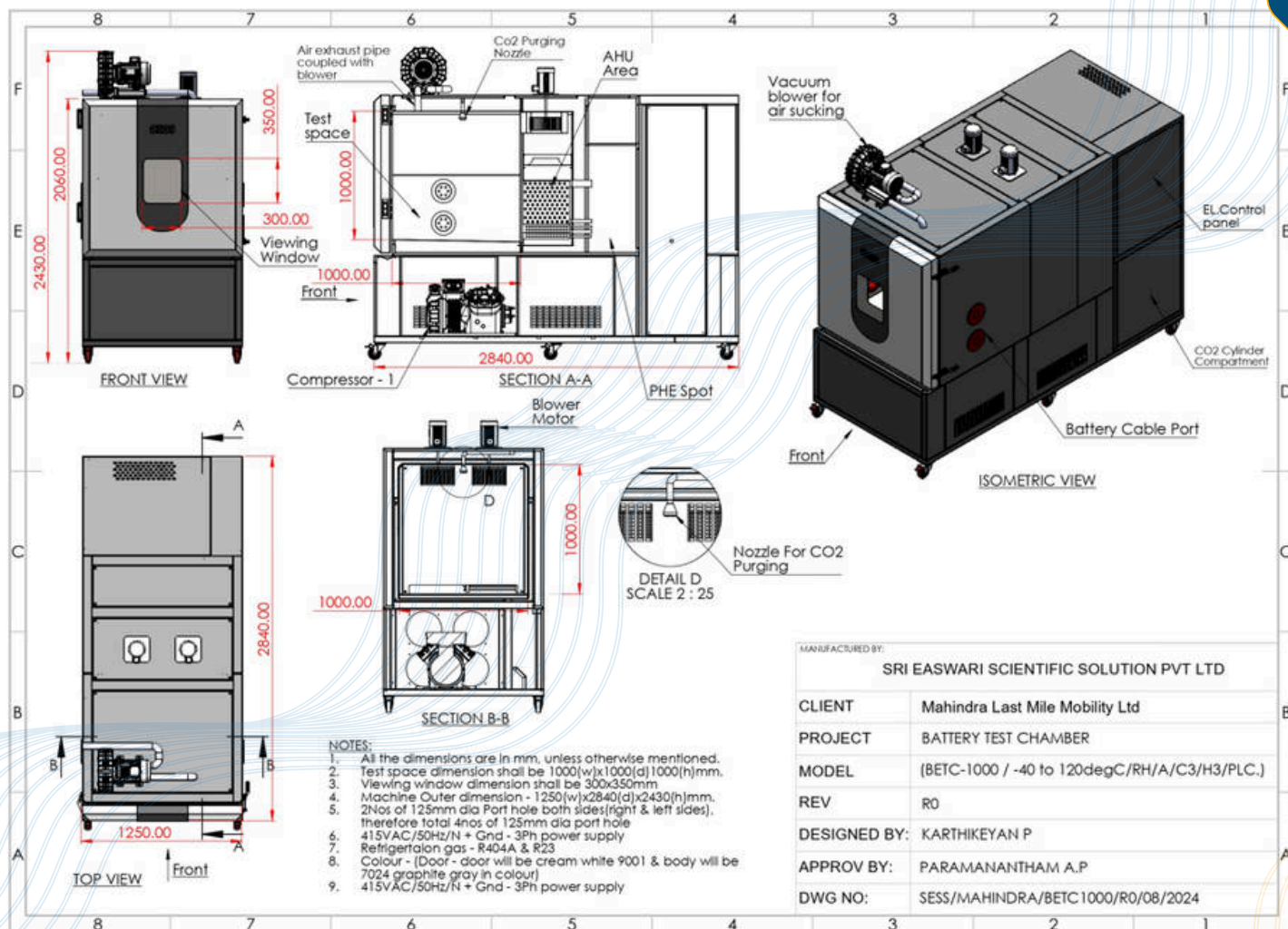
Electrical Connection

Specification: 3 Phase + Neutral + Earth, 415Vac, 50Hz

Sound Pressure Level:

Level: Less than 70 dB (A) at a distance of 1 meter from the front door





EUCAR Hazard Levels

These levels help assess the danger associated with handling batteries. They are defined by the European Council for Automotive R&D (EUCAR).

The levels range from 0 to 7, with increasing severity:

Level 0: No effect, no loss of functionality.

Level 1: Passive protection activated; cell reversibly damaged.

Level 2: Cell damaged irreversibly; replacement needed.

Level 3: Leakage, loss of mass < 50% (no fire or explosion).

Level 4: Leakage, loss of mass > 50% (no fire or explosion).

Level 5: Fire or flame (no rupture or explosion).

Level 6: Rupture (no explosion, but flying parts).

Level 7: Explosion (disintegration of the cell).

Based on your Offered test chamber hazard level it will applicable. Battery Test Chambers included with battery safety testing features equipped to handle up to hazard level 7:

- Emergency Stop Button
- High/Low Temperature Limit
- Rapid Temperature Change
- Audible and Visual Alarm
- Temperature Limited Sheath Heaters
- Enhanced Remote Door Lock
- Burst Disk/Low Flow Vent
- Safety Purge
- Product Sensors
- Reinforced Inner Wrap
- Chamber Alarm Input
- Chamber Alarm Output

1. Hydrogen (H₂)

Lower Explosive Limit (LEL): 4.0% (40,000 ppm)

Recommended Maximum Safety Level:

Alarm Threshold: 1,000 ppm (0.1%)

Shutdown Threshold: 2,000 ppm (0.2%)

Why: Hydrogen is extremely flammable, and setting the alarm and shutdown thresholds well below the LEL ensures that any significant hydrogen accumulation is detected and addressed before it can reach explosive levels.

3. Carbon Dioxide (CO₂)

Normal Atmospheric Level: ~400 ppm

Recommended Maximum Safety Level:

Alarm Threshold: 1,000 ppm to 2,500 ppm

Shutdown Threshold: 5,000 ppm (0.5%)

Why: Elevated CO₂ levels can indicate battery degradation or a potential fire risk. Setting these thresholds helps detect and mitigate risks early.

Chamber Construction: The chamber is meticulously crafted with a double-walled, insulated design employing argon Tig/arc welded stainless steel sheet (304) measuring 1.5/1.2 mm in thickness for the interior, while the outer walls are constructed using CRCA sheet of 1.5 mm thickness or As per the Tender RFQ. The outer body is manufactured using CNC technology, guaranteeing precision in production and featuring impeccably smooth joints without any protruding fixtures.

The chamber's door is coated with powder in RAL 9001 cream white, while the main chamber body is finished with RAL 7024 graphite grey powder coating. The chamber is ingeniously designed in a vertical trolley mounting configuration. It showcases a fully frontal opening double-walled insulated door, thoughtfully equipped with an exclusively sourced high-quality silicone rubber gasket. To ensure both security and insulation, the door is outfitted with robust hinges and a toggle-type locking mechanism.

This attention to construction details not only underscores the chamber's durability and functionality but also its aesthetic appeal.

2. Oxygen (O₂)

Normal Atmospheric Level: 20.9%

Recommended Safety Levels:

Minimum Safety Level: 19.5%

Maximum Safety Level: 23.5%

Why: Oxygen supports combustion, so maintaining levels within this range prevents hypoxia (low levels) and reduces fire risk (high levels). Alarms should trigger if levels drop below or rise above these thresholds.

4. Carbon Monoxide (CO)

Occupational Exposure Limit (OSHA): 50 ppm (for 8-hour exposure)

Recommended Maximum Safety Level:

Alarm Threshold: 25-35 ppm

Shutdown Threshold: 50 ppm

Why: CO is a toxic gas, and its presence can be an early indicator of battery failure or fire.



In our climatic test chamber's air circulation system, we incorporate the following components and features:

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1.Fan/Blower: The chamber is equipped with a fan or blower featuring a continuous-duty rated motor. This motor efficiently circulates air within the chamber.

2.Conditioning Space: Inside the main chamber, there's a dedicated conditioning area with baffles, housing essential components like heaters and cooling coils. These elements play a crucial role in air conditioning.

3.Uniform Temperature: Conditioned air is introduced into the main chamber to ensure consistent temperature distribution throughout the workspace, maintaining a stable temperature environment during tests.

4.Laminar Airflow: The chamber is designed for laminar airflow, achieved using a fan motor and impeller with an appropriate capacity. This setup guarantees even distribution of temperature and humidity within the chamber, adhering to specified limits.

5.Motor Shaft: TC/ETC - The motor shaft used for chamber air circulation is constructed with SS 304 without any joints. The motor is positioned outside the chamber, while an extended-length SS 304 shaft drives the fans. This design ensures smooth and reliable operation while preserving the chamber's internal environment.

ATC/AETC - "The mechanical design of the chamber's air circulation motor is constructed with a sturdy shaft connected by a flexible coupling. The inner shaft is made of SS 304 material and is connected to the blower drum, which features a double vacuum seal mechanism."

6.Motor Winding: For high-temperature chambers, we utilize motor windings of H class, whereas for chambers operating below 130°C, F class motor windings are employed.

These features combine to create an efficient and precise air circulation system within the climatic test chamber, facilitating accurate and dependable test conditions for various experiments and applications.



Inconel-type Air heaters with PTFE wire and silicon end cap.

The chamber is equipped with sealed Inconel tube heaters to maintain a uniform temperature. These heaters are meticulously designed to ensure even heat distribution throughout the chamber. Heat input is regulated by solid-state relays, controlled through a PID controller programmer.

One notable advantage of inconel tube heaters is their exceptional ability to withstand high temperatures and corrosive environments. Inconel, known for its outstanding durability and heat resistance, is well-suited for prolonged use in challenging conditions.



Compared to stainless steel heaters, inconel heaters exhibit superior resistance to oxidation, corrosion, and high-temperature degradation. They are engineered to deliver reliable and efficient heat transfer without direct radiation onto the test item, ensuring a uniform heating process that doesn't adversely affect the test specimen.

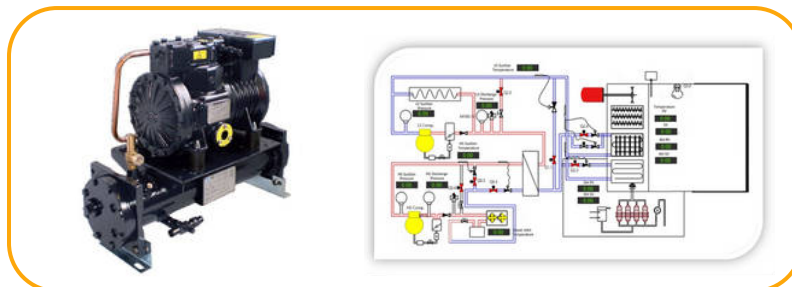
In conclusion, the use of sealed inconel tube heaters is highly recommended for maintaining precise and uniform chamber temperatures while ensuring the performance of the heating system.



For single-stage configurations, the system features either air or water-cooled refrigeration, incorporating a Hermetic compressor from Embraco, Danfoss, or Emerson, or a semi-hermetic compressor from Bitzer, Dorin, or Bock. These compressors operate with the environmentally friendly refrigerant R404A. The system includes essential safety features such as HP/LP cut-off switches, air inlet condensers/temperature protectors, backup fuse protection, expansion valves, and is engineered for continuous and trouble-free operation.

For double-stage cascade configurations, the system comprises either air or water-cooled refrigeration, equipped with a Hermetic compressor from Embraco or Danfoss, or a semi-hermetic compressor from Bitzer, Dorin, Bock, or Emerson. These compressors utilize environmentally friendly refrigerants R404A and R23. Safety features such as HP/LP cut-off switches, air inlet condensers/temperature protectors, backup fuse protection, expansion valves are included, ensuring continuous and trouble-free operation.

Our chamber's refrigeration system is designed for optimal performance, with variations in specifications based on different models.



Single-stage compressor for temperatures up to -40°C , and a double-stage compressor for temperatures below -70°C . We select our compressors from reputable brands like Bitzer, Dorin, Bock, Danfoss, SECOP, or Embraco.

Environmentally friendly refrigerants are used, with R404A for single-compressor systems and R404A/R23 for cascade systems.

Safety features include HP/LP cut-off switches, pressure transmitters for continuous monitoring, and suction line temperature sensors for enhanced safety.

The system can have a water or air-cooled condenser based on the model. Temperature monitoring and safety limits are integrated into our PLC system for effective control and protection.

The compressor and condenser unit are positioned below the chamber for optimal performance.

For condenser cooling, our PLC system monitors the air or water inlet temperature. If it exceeds 26°C (water) or 40°C (air), a safety alarm triggers, and chamber operation stops until the temperature returns to normal, which can be adjusted through HMI and PC software.

A separate control panel will be provided, attached to the main chamber, which will house the programmer, on/off switches, fuses, contactors, indicating lamps. Channel type wiring will be implemented using suitable copper wires with appropriate current ratings, along with marking ferrules, crimped dowel terminals, elmex connectors, and so on.

The electrical and instrumentation panel will be meticulously wired in compliance with international standards to ensure safety and reliability. Each electrical cable will be tagged with ID numbers, and the corresponding details will be clearly documented in the maintenance manual. Comprehensive circuit diagrams and relevant information will also be included.

To facilitate accurate and efficient design, we utilize SolidWorks 3D software for developing our drawings. This advanced software enables us to create detailed structural and electrical drawings that adhere to industry standards and specifications. By utilizing SolidWorks, we can ensure precise and comprehensive documentation of the electrical components and wiring layout of the Thermal cyclic test chamber.

Furthermore, we take pride in mentioning that our standard test chamber has been certified by TUV and carries the CE mark. This certification validates that our Stand test chamber only complies with the essential health, safety, and environmental protection requirements of the European Union. It serves as a testament to our commitment to providing high-quality, safe, and reliable equipment.

Our dedication to utilizing modern software tools and obtaining relevant certifications reflects our unwavering commitment to delivering superior products that meet the highest industry standards and exceed customer expectations.

Solid state relays for control for dry heater



In our PLC model, all chamber indicators are available on the soft screen of the HMI itself, along with a buzzer option for the following:

- SPP trip
- Air Circulation Fan motor OLP / Hot Zone Fan motor OLR & Cold Zone Fan motor OLR
- Over Temperature in Chamber inner / Over Temperature in Hot Zone & Under Temperature in Cold Zone
- HS Compressor OLR
- HS Compressor over temperature
- HS Compressor HP trip
- HS Compressor LP trip
- LS Compressor OLR
- LS Compressor over temperature
- LS Compressor HP trip
- LS Compressor LP trip
- a. During transfer of basket (missing of the basket)
- Air Heater protection etc....



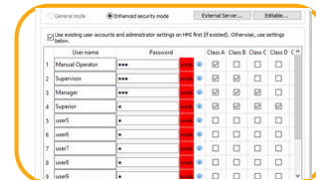
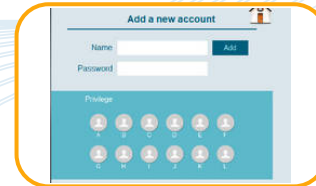


Data Storage and Profile Creation

The Table Top Climatic Test Chamber boasts extensive data storage capabilities, allowing for the retention of test data for up to 100 days. Additionally, users have the ability to create and save up to 100 test profiles, streamlining the setup process and facilitating efficient testing procedures.

Certainly, our HMI provides comprehensive access level password protection. End-users will have their login IDs with customizable permissions based on different levels. The system allows for the creation of up to 12 levels of passwords, each with its specific permissions.

Furthermore, users with appropriate access levels can edit high/low alarm limits directly through the HMI interface. This functionality ensures that users can tailor alarm settings to their specific requirements without compromising security or ease of use.



Humidity System

Our Climatic only for ETC model test chamber is equipped with an immersion heater-based boiler tank. The chamber includes a 15-20 liter inbuilt water storage tank, which is connected to the boiler tank through a water solenoid valve. The system features an automated water filling system with an electrical water level sensor that maintains the water level in the boiler tank. This ensures efficient and continuous humidity control inside the test chamber.

Dehumidification System

The dehumidification system is controlled via PLC logic. It operates through a de-humidity solenoid valve, integrated with the refrigeration system. The cooling coil contains a refrigeration cold tube that absorbs moisture from the chamber's air. The system monitors the cold tube temperature through the PLC. When dehumidification is required, the cold tube captures the moisture, freezes it into water particles, and drains the condensed water, effectively reducing the chamber's internal humidity.

Certification





THANK YOU !



Presented By
SESS TECHNICAL TEAM

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